FINAL REVIEW · OCTOBER 2026

Demeure AI

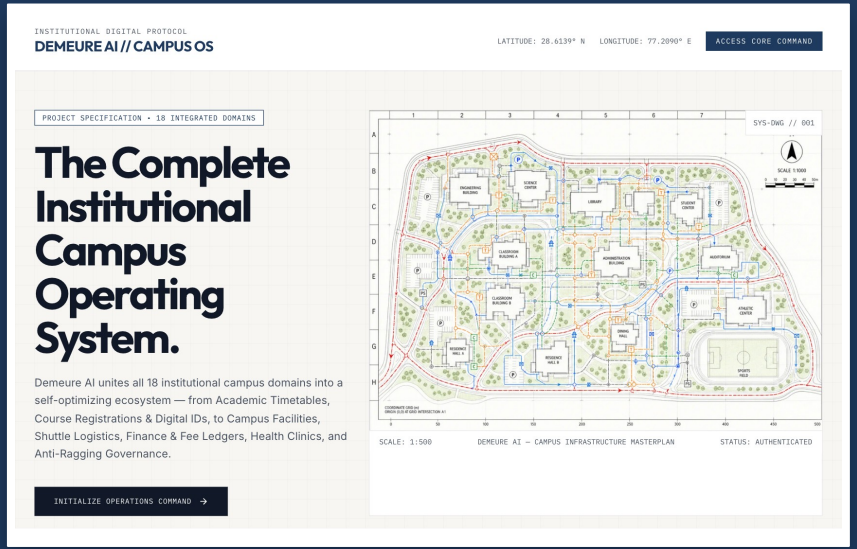
Intelligent campus operating system for hostel and campus life

TEAM

Pratik Kumar · 24BML0027
Thakur Chand Choudhary · 24BML0012
Nikunj Goydani · 24BEC0338

FACULTY GUIDE

Dr. Rajalaxmi S
Assistant Professor Sr. Grade 1
School of Electronics Engineering (SENSE), VIT Vellore



Live product screenshots throughout

Campus operations still run on paper

Paper registers

Leave and visitor passes signed on paper; no audit trail for wardens or parents.

Lost complaints

Maintenance issues buried in chat threads with no priority or owner.

Unused surveys

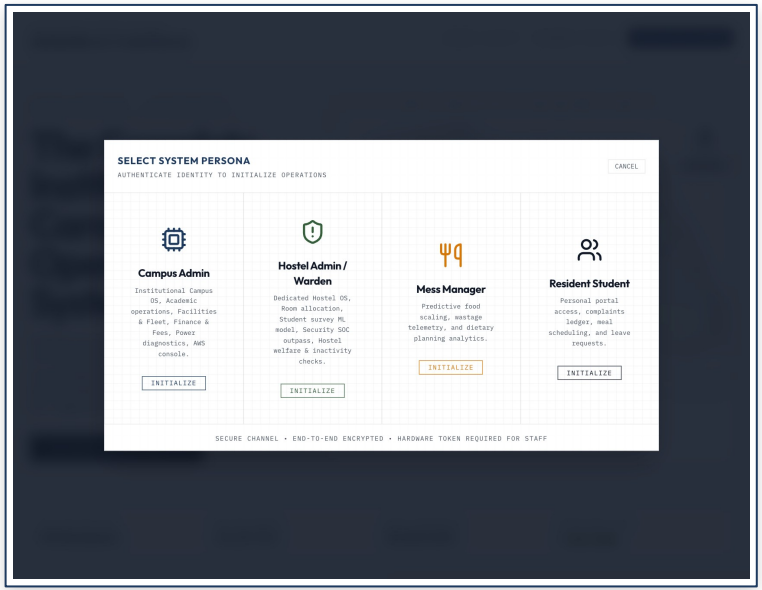
Roommate preferences collected in spreadsheets, then allocated by hand.

Manual billing

Electricity and mess dues calculated by hand and paid by screenshot.

Result: slow approvals, unsafe incidents missed, and no single source of truth.

One platform, four roles, one source of truth



Persona picker on the landing page

Campus Admin
Finance, facilities, power billing, AWS hub

Hostel Warden
Residents, rooms, surveys, gate, welfare

Mess Manager
Weekly menu publishing, food feedback

Resident Student
Dues, leave, visitors, complaints, AI help

What ships today

14

operational modules across hostel and campus

4

role-scoped personas with server-side access control

100

residents with encrypted personal records

41

automated API and domain tests

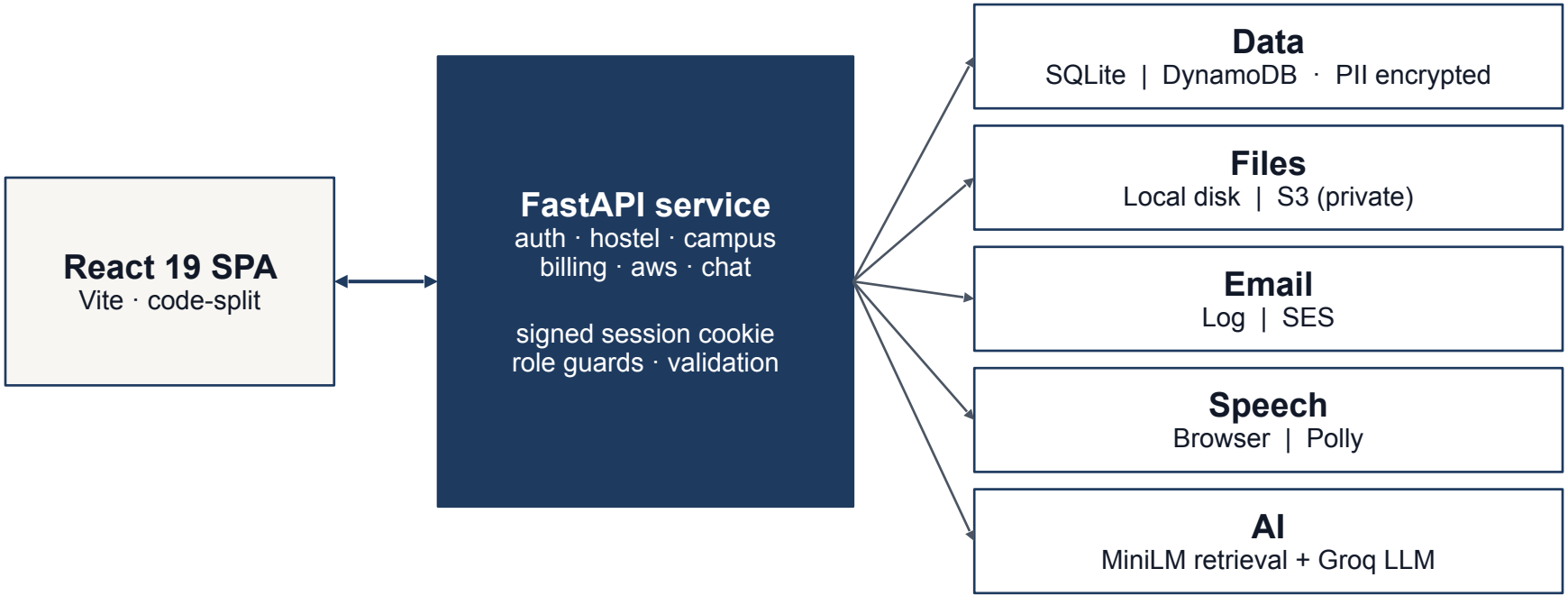
57

browser checks across 4 roles, all passing

76 kB

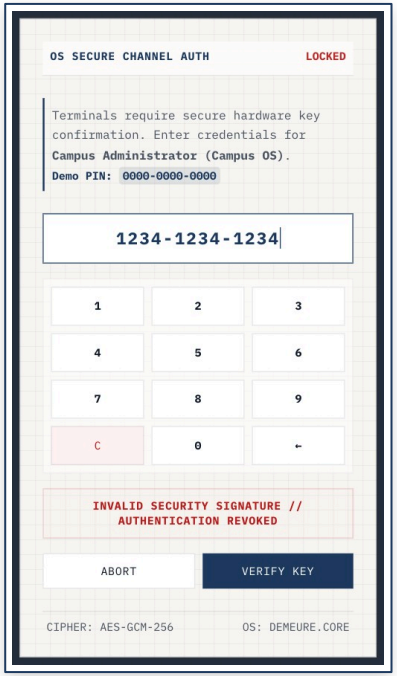
first-load JavaScript (gzip), down from 386 kB

Hybrid by design: runs locally or on AWS

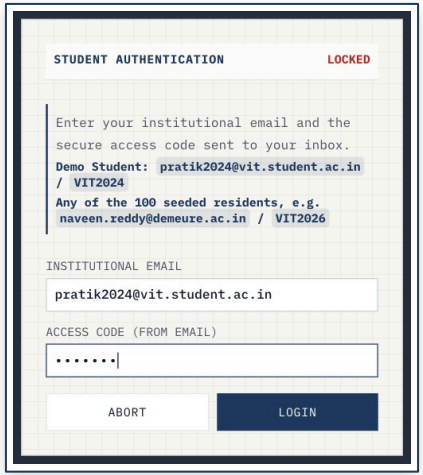


Same code, two modes: no AWS keys means durable local mode; add keys and it switches to AWS.

Every sign-in is checked by the server



Wrong PIN rejected



Student sign-in

Server-verified
PBKDF2-hashed PINs and codes; signed, expiring sessions

Role-scoped APIs
A student calling an admin endpoint gets 403

Brute-force throttle
10 failures in 5 minutes blocks the client

Safe by default
Demo hints only in dev; prod requires real secrets

USE-CASE SCENARIOS

Twelve flows, captured live

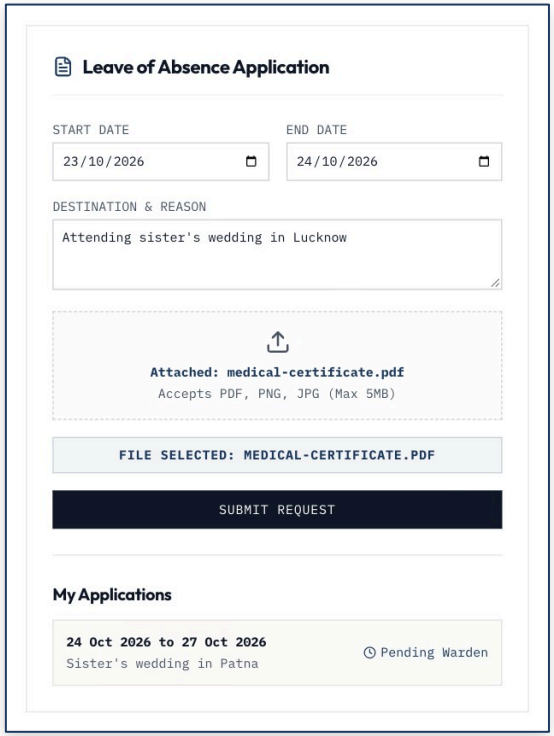
Each scenario crosses roles: a student acts, staff respond, and both see the same persisted record.

Leave request with medical proof

1. **Fill dates and reason**
   Past dates and end-before-start are blocked

2. **Attach the certificate**
   PDF/PNG/JPG, checked by content, kept private

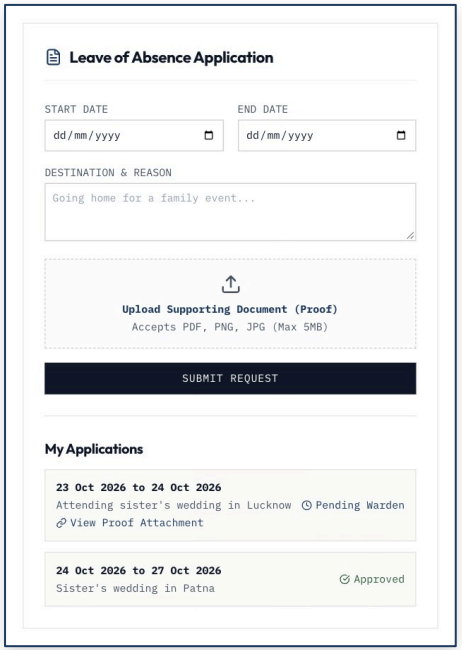
3. **Submit**
   Guardian is emailed automatically (SES)



Warden reviews the proof and approves

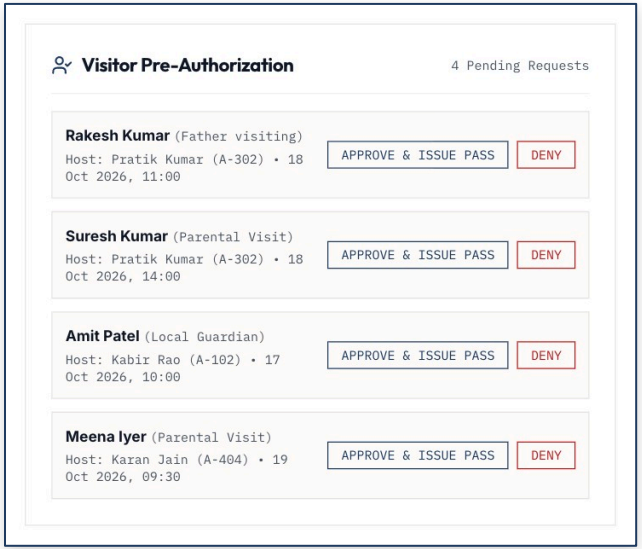


Pending queue with the private attachment link

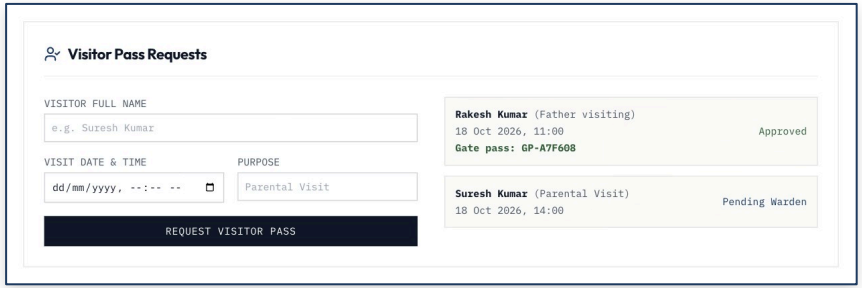


The student sees "Approved" instantly

Visitor pass in two clicks



Warden approves the pending request



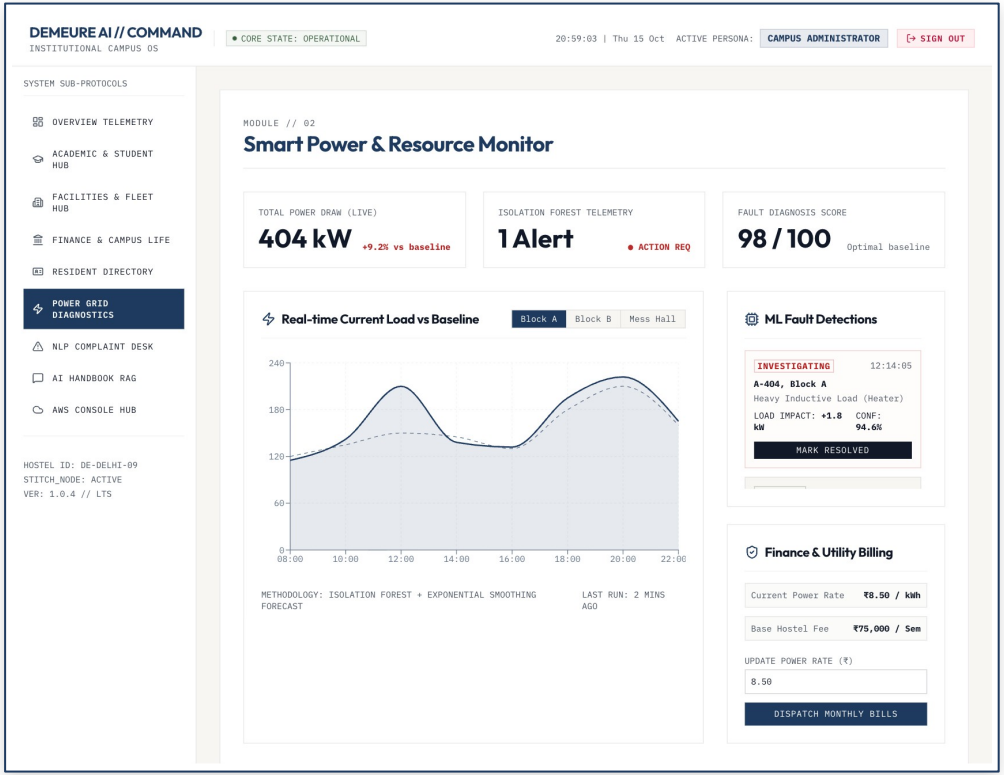
Student receives a gate pass code on approval

New billing cycle from the power monitor

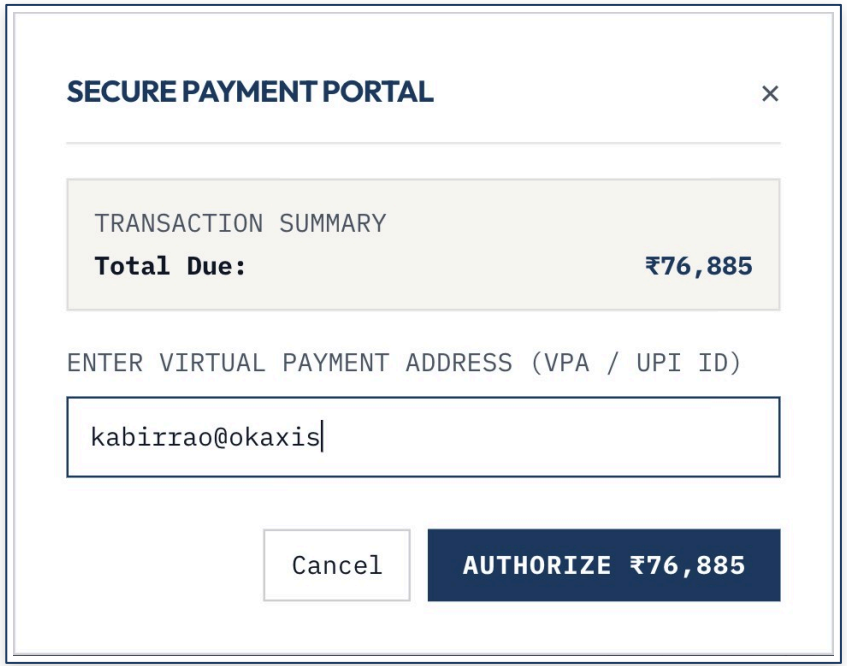
1. **Inspect the load**
   Per-zone consumption vs. baseline

2. **Set the rate**
   Validated between ₹0.01 and ₹100 per kWh

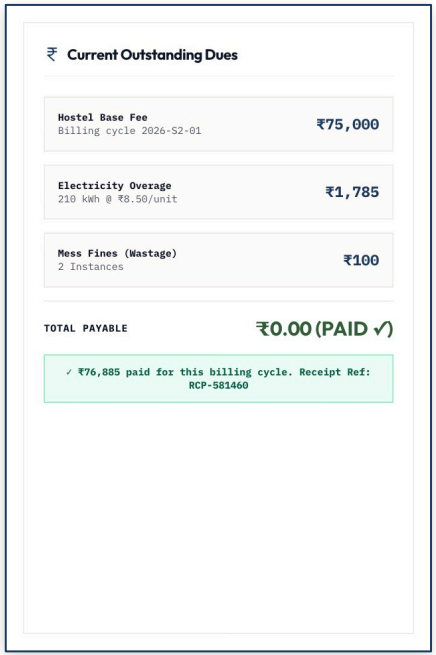
3. **Dispatch**
   A new cycle ID is pushed to every student portal



Dues calculated on the server, paid by UPI

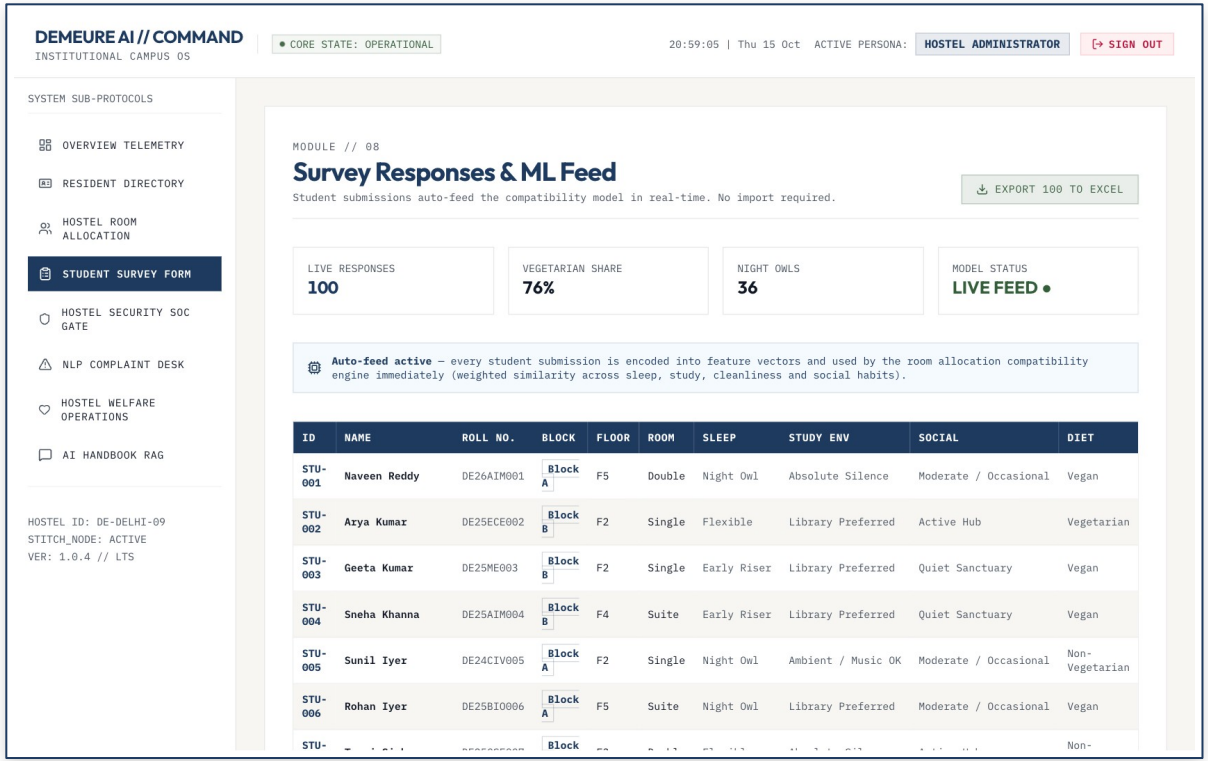


UPI authorisation for the server-computed total



Receipt stored against the billing cycle

100 seeded students feed the allocation engine



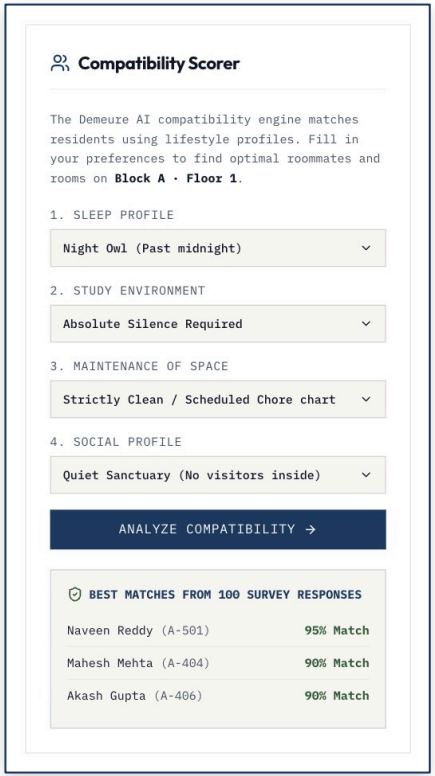
100

survey responses from the seeded residents

1-click

Excel export with the ML feature matrix

Compatibility ranks real roommates

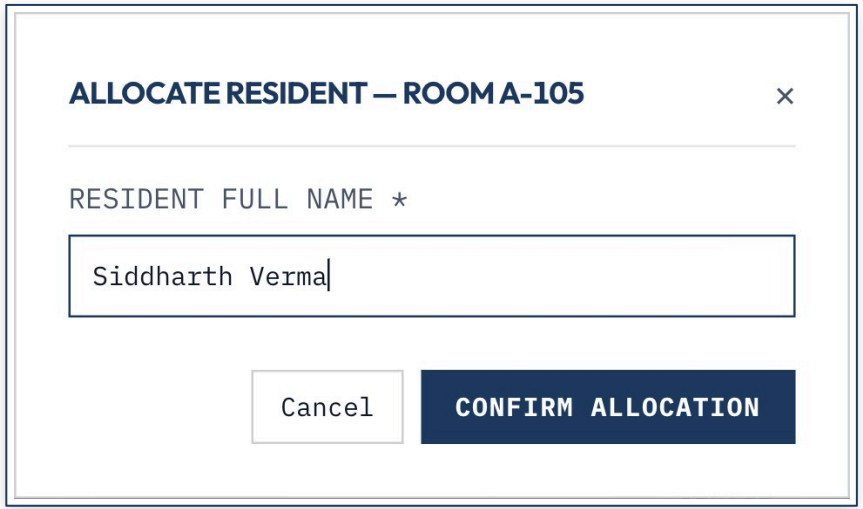


Similarity weights used for every match

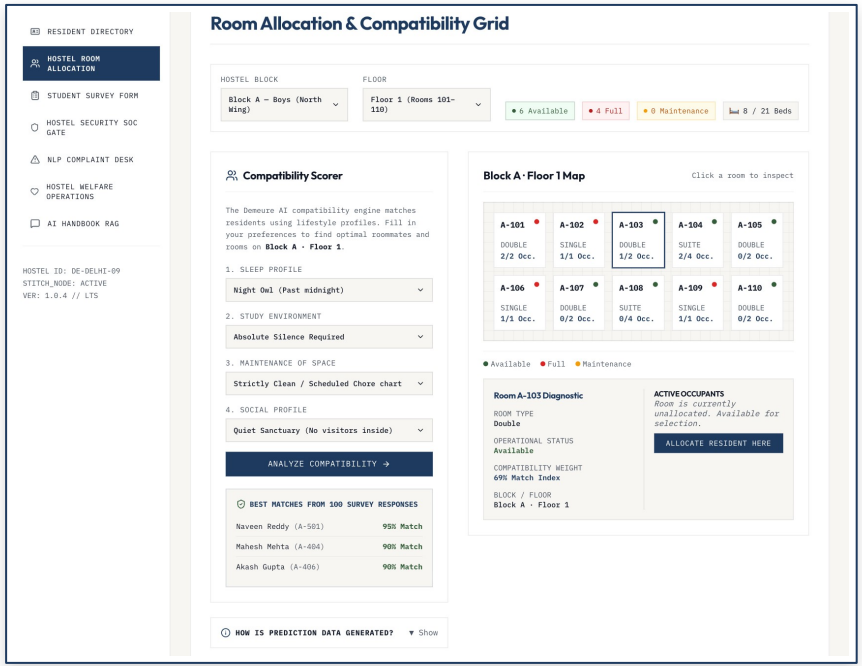


Search a Night Owl / silent / strictly clean / quiet profile in Block A: the best residents score 95%, 90% and 90%. Single-gender blocks only rank residents of that block.

Capacity-checked room allocation

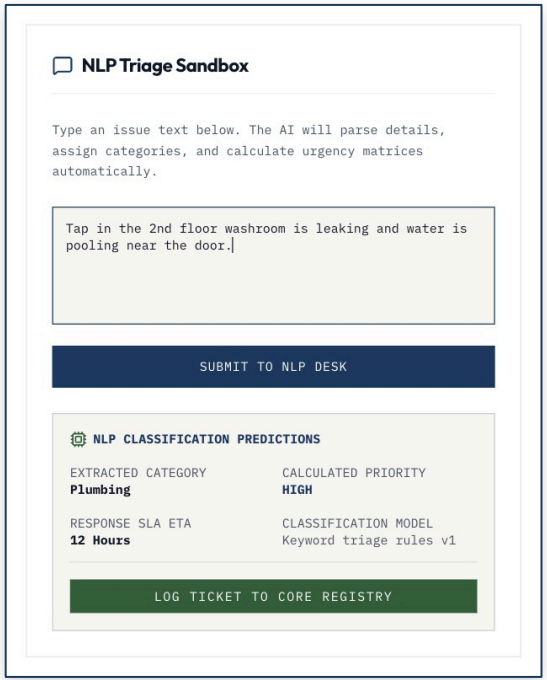


Allocating a resident to a room with a free bed

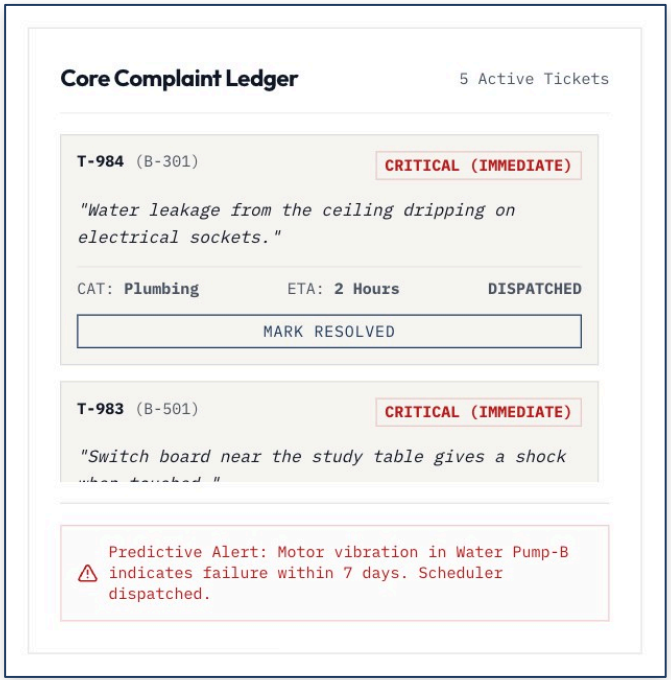


Occupancy updates and persists

Complaints triaged by severity



Washroom leak: Plumbing, high priority, 12 h SLA



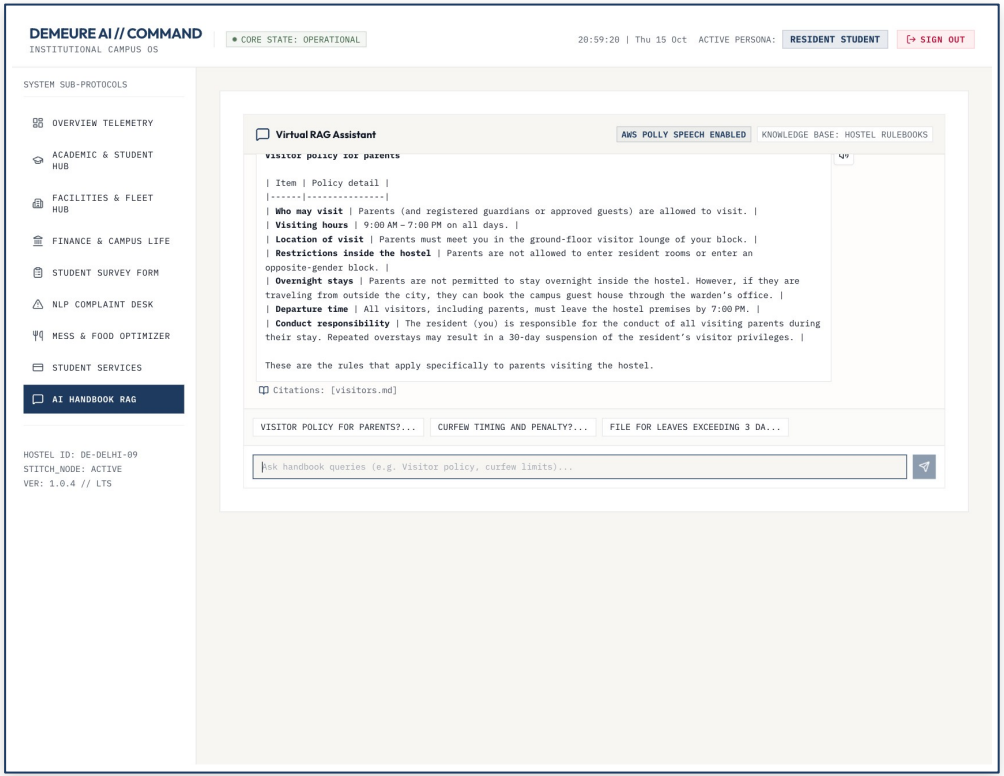
Seeded hazards escalated to critical, 2 h target

Handbook assistant answers with citations

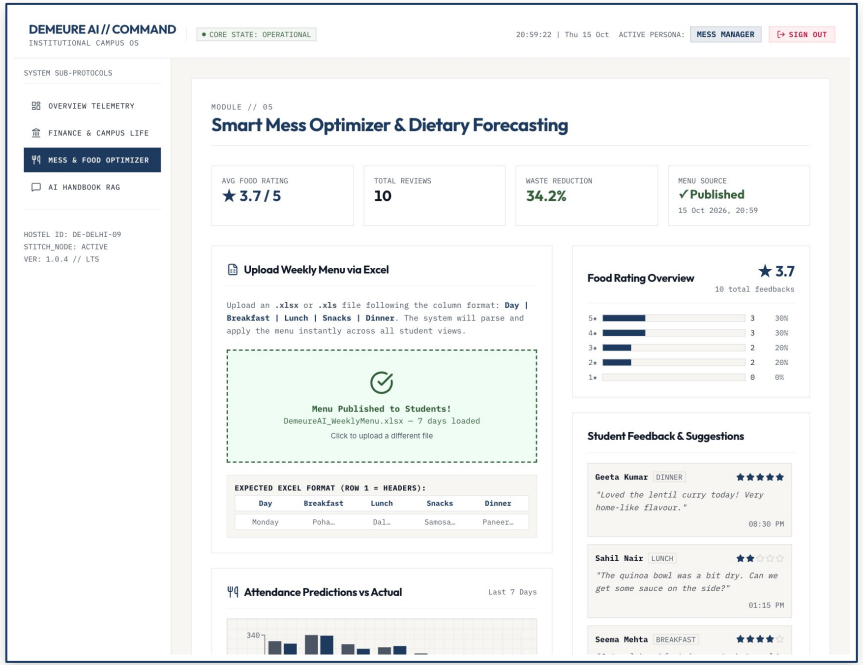
1. **Retrieve**
   MiniLM embeddings over eight rulebooks

2. **Ground**
   LLM answers only from the retrieved text

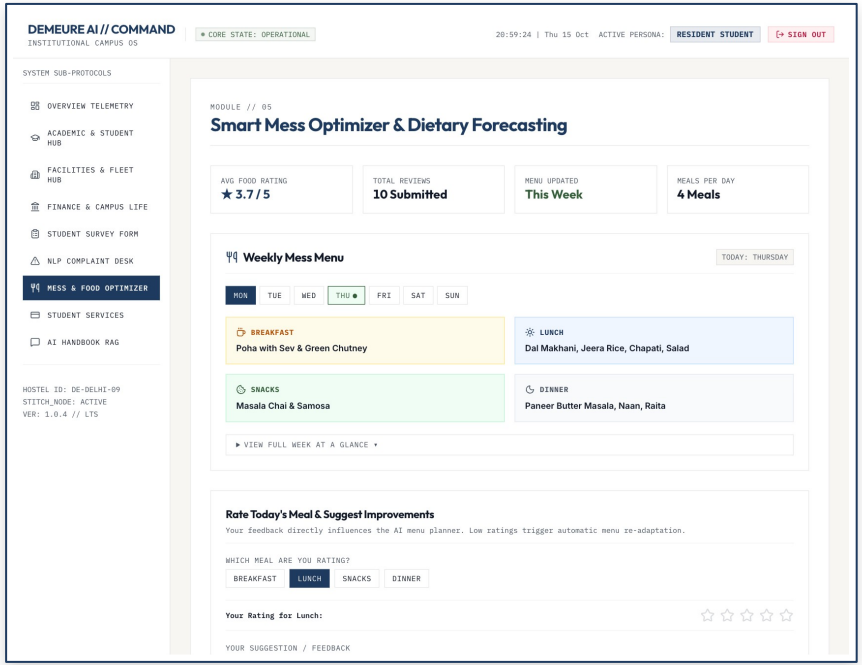
3. **Cite and speak**
   Source files listed; Polly or browser voice



Weekly menu published from Excel

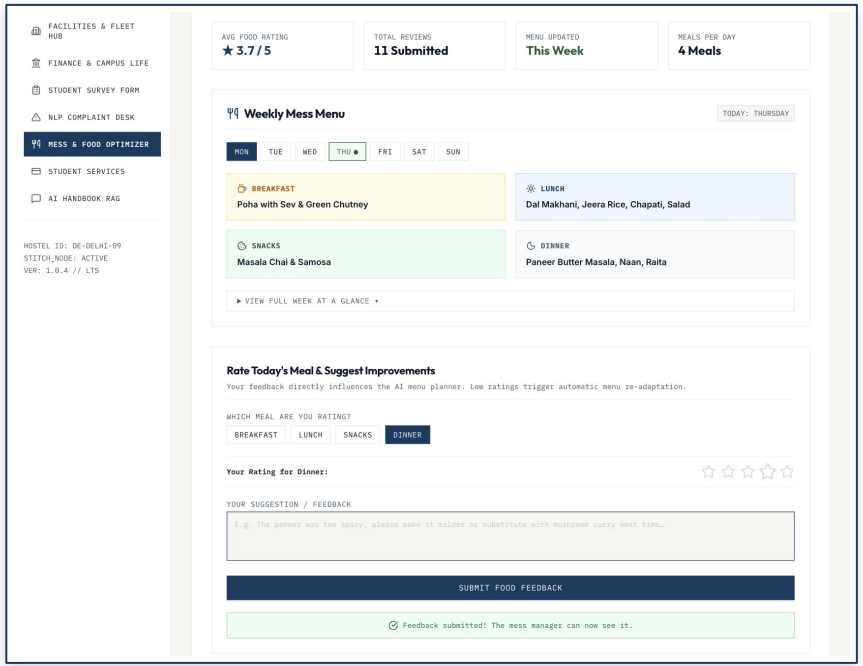


Upload and publish the weekly menu

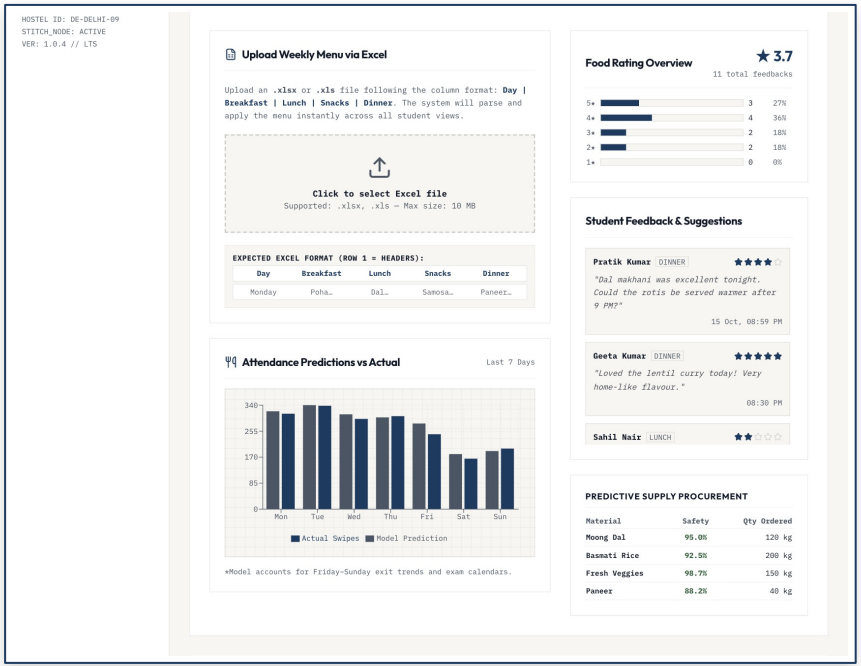


Students see the published menu

Student ratings reach the kitchen

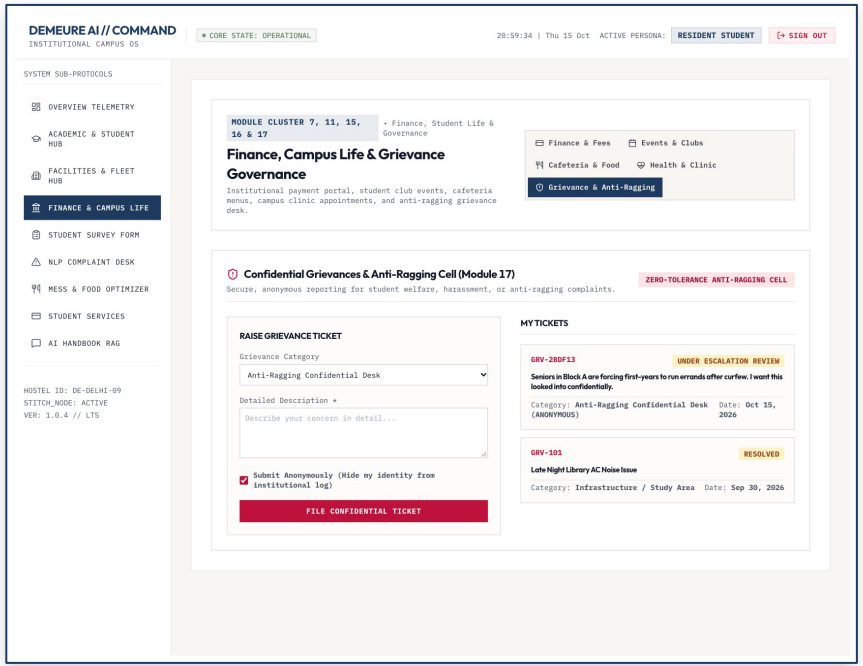


Student rates dinner and suggests a change

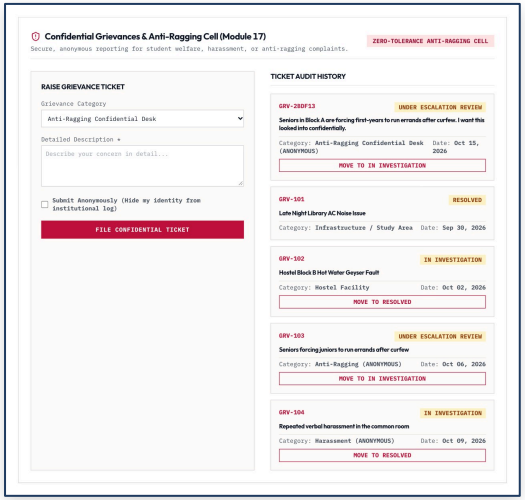


The manager sees it in the feedback feed

Anonymous grievances, accountable process

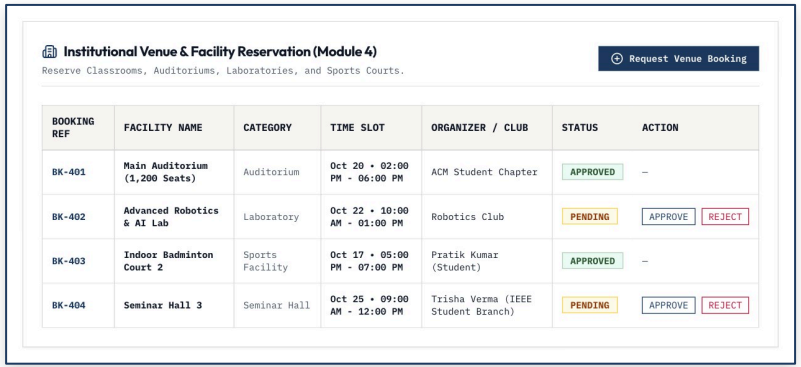


Student files an anti-ragging ticket anonymously

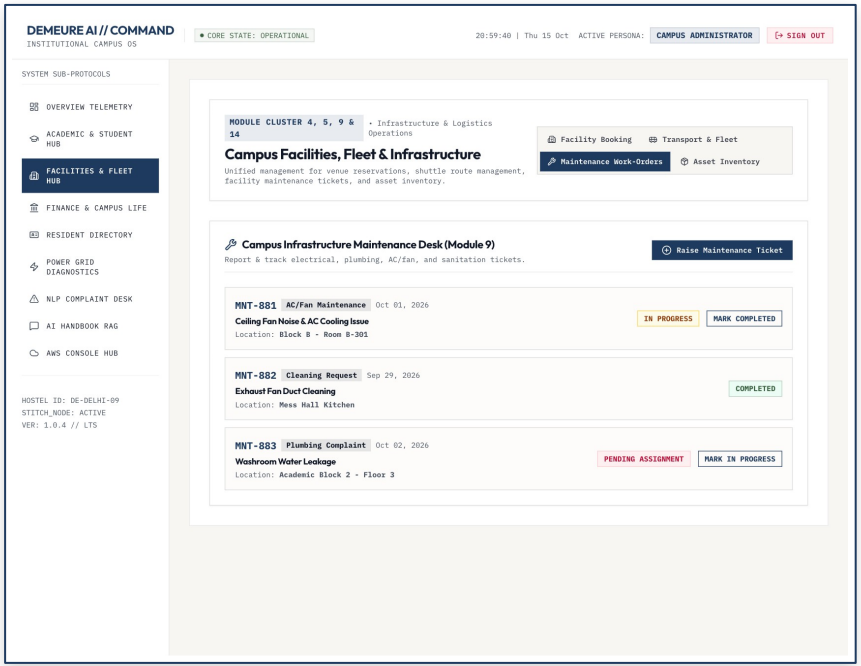


Admins triage it without seeing the filer

Facility bookings and work orders

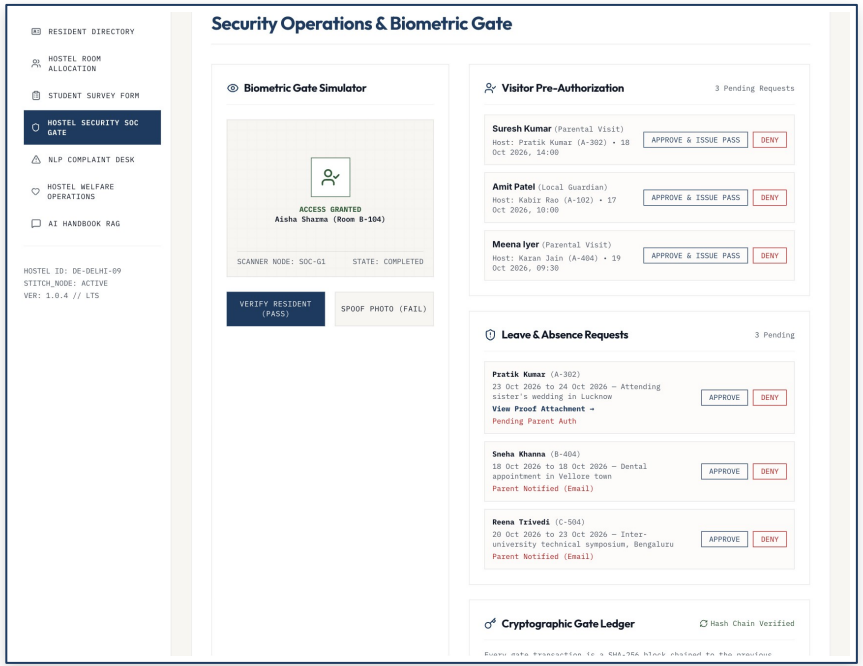


Admin approves or rejects venue bookings



Work orders move from pending to completed

Gate security with a tamper-evident ledger

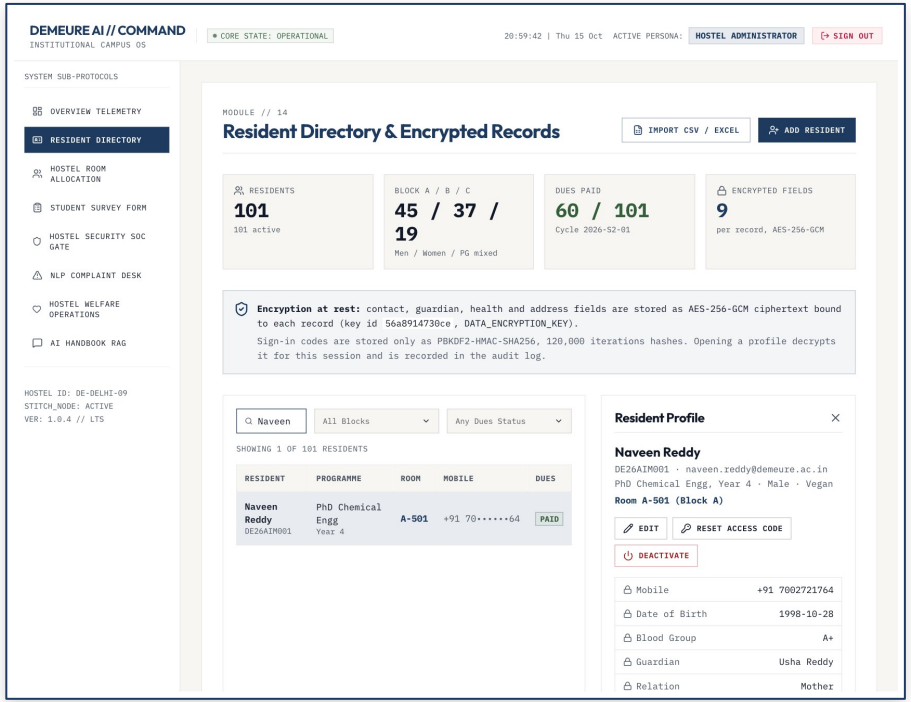


Resident verified at gate SOC-G1

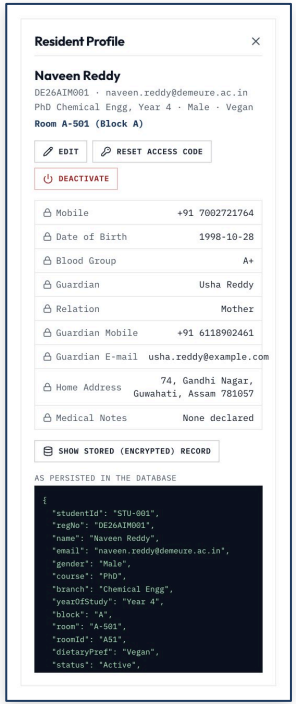


Hash chain verified on every read

100 residents, personal data encrypted



Directory with a decrypted profile (audited)



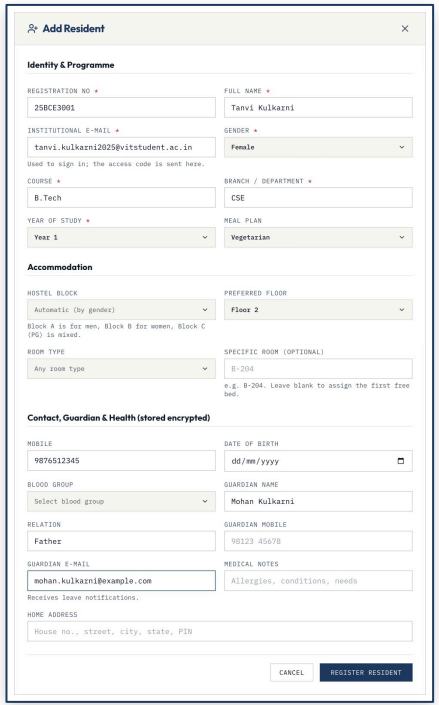
As stored

Encrypted at rest
9 personal fields per record, AES-256-GCM

Bound to the record
Ciphertext moved to another record fails

Masked and audited
Full view logged; codes stored as hashes

Register one resident or import a whole batch



Validated registration form

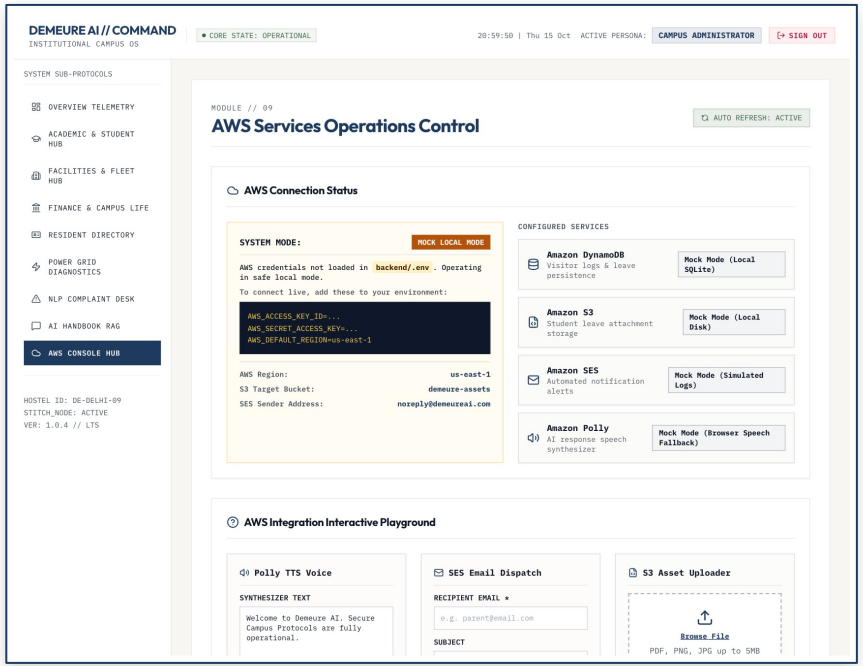


CSV import: 2 registered, 1 row reported

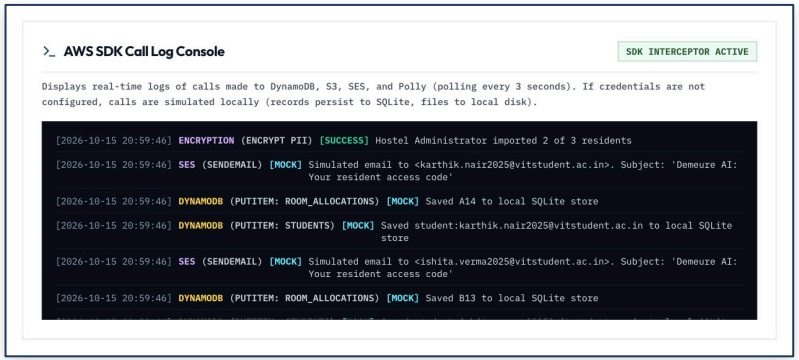
Bed assigned automatically
By block, floor and room type; never mixes genders in a room

One-time access code
E-mailed to the resident; stored only as a PBKDF2 hash

AWS hub: live or mock, always observable



Service status and test console



Every write, email and upload in the log feed

PRODUCTION READINESS

From prototype to deployable

What changed under the hood to make Demeure safe to run for a real campus.

Built to be operated, not just demoed

Real authentication
Hashed secrets, signed sessions, throttling

Role guards
Every endpoint checks the caller's role and ownership

Encrypted records
AES-256-GCM personal fields; SQLite or DynamoDB

Private files
Type, content and size checks; owner/staff access only

Secure headers
CSP, HSTS, frame-deny; docs hidden in production

Tested
41 API tests, 57 browser checks, readability audit

CI pipeline
Lint, build, audit and tests on every push

One-step deploy
Single Docker image, health check, data volume

Where Demeure goes next

Live payments

Razorpay or PayU UPI gateway in place of the simulated collection

SIS integration

Real timetables, grades and library holdings

Gate hardware

Connect real biometric scanners to the ledger API

Campus SSO

Google Workspace or Azure AD sign-in for staff and students

THANK YOU

Demeure AI

Questions, and a live walkthrough of any scenario.

